



Induction Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 Use mathematical induction to prove that $n^3 + 2n$ is divisible by 3 for all positive integers n . 3

Prove true for $n = 1$

$1^3 + 2 \times 1 = 3$ which is divisible by 3 so true for $n = 1$ ☒

Assume true for $n = k$

$$\frac{k^3 + 2k}{3} = M \text{ (where } M \text{ is a positive integer)}$$

$$k^3 = 3M - 2k$$

Prove true for $n = k + 1$

RTP that $(k+1)^3 + 2(k+1)$ is divisible by 3

$$(k+1)^3 + 2(k+1) = k^3 + 3k^2 + 3k + 1 + 2k + 2$$

$$= k^3 + 3k^2 + 5k + 3$$

$$= 3M - 2k + 3k^2 + 5k + 3 \text{ (from assumption)} \quad \checkmark$$

$$= 3(M + K + 1) \rightarrow \text{(divisible by 3)} \quad \checkmark$$

true for $n = k + 1$, proved by mathematical induction

- 2 Prove, by mathematical induction, that for all integers $n \geq 1$, 3

$$1 + 2 \times 2^{-1} + 3 \times 2^{-2} + 4 \times 2^{-3} + \dots + n \times 2^{-(n-1)} = \frac{2^{n+1} - n - 2}{2^{n-1}}$$

$$1 + 2 \times 2^{-1} + 3 \times 2^{-2} + 4 \times 2^{-3} + \dots + n \times 2^{-(n-1)} = \frac{2^{n+1} - n - 2}{2^{n-1}}$$

$$\text{For } n = 1 \quad \text{LHS} = 1 + 2 \times 2^{-1} = 1 \quad \text{RHS} = \frac{2^{1+1} - 1 - 2}{2^{1-1}} = 1$$

$\therefore$ Statement is true for $n = 1$

Assume statement true for $n = k$, that is,

$$1 + 2 \times 2^{-1} + 3 \times 2^{-2} + 4 \times 2^{-3} + \dots + k \times 2^{-(k-1)} = \frac{2^{k+1} - k - 2}{2^{k-1}}$$

Prove true for $n = k + 1$ That is, we show that

$$1 + 2 \times 2^{-1} + 3 \times 2^{-2} + 4 \times 2^{-3} + \dots + k \times 2^{-(k-1)} + (k+1) \times 2^{-(k+1-1)} = \frac{2^{k+1+1} - (k+1) - 2}{2^{k+1-1}}$$

$$\text{LHS} = \frac{2^{k+1} - k - 2}{2^{k-1}} + (k+1) \times 2^{-(k+1-1)}$$

$$= \frac{2^{k+1} - k - 2}{2^{k-1}} + (k+1) \times 2^{-1(k+1-1)}$$

$$= \frac{2^{k+1} - k - 2}{2^{k-1}} + \frac{k+1}{2^k}$$

$$= \frac{2^{k+2} - 2k - 4}{2^k} + \frac{k+1}{2^k} \quad * (\times \frac{2}{2})$$

$$= \frac{2^{k+2} - k - 3}{2^k} = \text{RHS}$$

$\therefore$ By the principle of mathematical induction the statement is true for $n \geq 1$

3 Use the process of Mathematical Induction to prove:

3

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n+1)!$$

for all positive integers n .

step 1: Prove true for $n=1$

$$\begin{array}{lll} \text{LHS} = 2 \times 1! & \text{RHS} = 1(1+1)! & \\ = 2 & = 2! & \therefore \text{LHS} = \text{RHS} \\ & = 2 & \text{True for } n=1 \end{array}$$

step 2: Assume true for $n=k$

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k! = k(k+1)! \quad *$$

step 3: Prove true for $n=k+1$

RTP: $\underbrace{2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k!}_{S_k} + [(k+1)^2 + 1](k+1)! = (k+1)(k+2)!$

$$\begin{aligned} \text{LHS} &= k(k+1)! + [k^2 + 2k + 1 + 1](k+1)! \quad (\text{by assumption}) \\ &= (k+1)! [k + k^2 + 2k + 2] \\ &= (k+1)! [k^2 + 3k + 2] \\ &= (k+1)! (k+2)(k+1) \\ &= (k+2)! (k+1) \\ &= \text{RHS} \end{aligned}$$

step 4: Since true for $n=1$, then true for $n=2, n=3$ and so on for all positive integers.

4 Use the principle of mathematical induction to show that for all integers $n \geq 1$,

3

$3 \times 5^{2n+1} + 2^{3n+1}$ is divisible by 17.

Required to prove: $3 \times 5^{2n+1} + 2^{3n+1}$ is divisible by 17.

Consider $P(1)$.

$$3 \times 5^{2(1)+1} + 2^{3(1)+1} = 391 = 17 \times 23$$

$\therefore$ true for $n=1$

Assume $P(k)$ true, for $k \in \mathbb{Z}^+$

i.e. $3 \times 5^{2k+1} + 2^{3k+1} = 17M$, where M is some integer.

Prove $P(k+1)$ true.

$$\begin{aligned} \text{i.e. } 3 \times 5^{2(k+1)+1} + 2^{3(k+1)+1} & \\ &= 3 \times 5^{2k+3} + 2^{3k+4} \\ &= 3 \times 5^2 \times 5^{2k+1} + 2^3 \times 2^{3k+1} \\ &= 3 \times 25 \times 5^{2k+1} + 25 \times 2^{3k+1} - 17 \times 2^{3k+1} \\ &= 25(3 \times 5^{2k+1} + 2^{3k+1}) - 17 \times 2^{3k+1} \end{aligned}$$

Now using assumption,

$$\begin{aligned} &= 25(17M) - 17 \times 2^{3k+1} \\ &= 17(25M - 2^{3k+1}), \text{ where } 25M - 2^{3k+1} \text{ is an integer.} \end{aligned}$$

$\therefore P(k+1)$ is true if $P(k)$ is true.

- 5 Use mathematical induction to prove that $3^{2n+1} + 2^{n-1}$ is divisible by 7, for all positive integers n . 3

n .

Let $P(n)$ be the proposition
[3] that $3^{2n+1} + 2^{n-1}$ is divisible by 7
for all integers $n > 0$.

Prove that $P(1)$ is true

$$\therefore \text{For } n=1, 3^{2 \cdot 1 + 1} + 2^{1-1} = 3^3 + 2^0 = 28$$

Since 28 is divisible by 7,
 $P(1)$ is true.

Assume that $P(k)$ is true. That is,

$$3^{2k+1} + 2^{k-1} = 7p, p \text{ is an integer}$$

$$\therefore 2^{k-1} = 7p - 3^{2k+1}$$

$$\frac{2^k}{2} = 7p - 3^{2k+1}$$

$$\therefore 2^k = 2(7p - 3^{2k+1}) *$$

Prove that $P(k+1)$ is true. That is

$$3^{2(k+1)+1} + 2^{(k+1)-1} = 7q, q \text{ integer}$$

$$\therefore \text{LHS} = 3^{2(k+1)+1} + 2^{(k+1)-1}$$

$$= 3^{2k+3} + 2^k$$

$$= 3^{2k+3} + 2 \times 7p - 2 \times 3^{2k+1}$$

$$= 9 \times 3^{2k+1} + 2 \times 7p - 2 \times 3^{2k+1}$$

$$= 7 \times 3^{2k+1} + 7 \times 2p$$

$$= 7(3^{2k+1} + 2p)$$

$$= 7q, q \text{ is an integer}$$

$$= \text{RHS}$$

$\therefore$ If $P(k)$ is true, $P(k+1)$ is also true

$\therefore$ The proposition $P(n)$ is true
by the principle of Math. Ind.
for all $n > 0$.

- 6 Prove by mathematical induction that $2^{2n} + 15n - 1$ is divisible by 9 for $n \geq 1$. 3

Prove true for $n=1$

$$2^{2(1)} + 15(1) - 1 = 4 + 15 - 1 = 18 \text{ which is divisible by 9.}$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$\therefore 2^{2k} + 15k - 1 = 9M \quad (M \text{ is an integer})$$

- induction hypothesis

Prove true for $n=k+1$

$$\begin{aligned} 2^{2(k+1)} + 15(k+1) - 1 &= 2^2 \cdot 2^{2k} + 15k + 14 \\ &= 4 \cdot 4^k + 15k + 14 \\ &= 4(9M - 15k + 1) + 15k + 14 \\ &= 9(4M) - 45k + 18 \\ &= 9(4M - 5k + 2) \end{aligned}$$

which is divisible by 9

$\therefore$ by principle of induction $2^{2n} + 15n - 1$ is
divisible by 9.

$$\cos\theta + \cos 3\theta + \cos 5\theta + \dots + \cos(2n-1)\theta = \frac{\sin 2n\theta}{2\sin\theta}$$

where $\sin\theta \neq 0$ and $n \in \mathbb{Z}^+$.

(i) Case $n=1$: $2n-1=1$.

$$RHS = \frac{\sin 2\theta}{2\sin\theta} = \frac{2\sin\theta \cos\theta}{2\sin\theta} = \cos\theta = LHS.$$

So true for $n=1$.

Suppose it's true for $n=k$,

$$i.e., \sum_{i=1}^k \cos((2i-1)\theta) = \frac{\sin 2k\theta}{2\sin\theta}.$$

$$\begin{aligned} \text{Then } \sum_{i=1}^{k+1} \cos((2i-1)\theta) &= \frac{\sin 2k\theta}{2\sin\theta} + \cos((2k+1)\theta) \quad (\text{from assumption}) \\ &= \frac{\sin 2k\theta}{2\sin\theta} + \frac{2\sin\theta \cos(2k\theta + \theta)}{2\sin\theta} \\ &= \frac{\sin 2k\theta + \sin 2k\theta + \sin(2k\theta + 2\theta)}{2\sin\theta} \quad (\text{since } \sin A \cos B = \frac{1}{2}(\sin(A+B) - \sin(A-B))) \\ &= \frac{\sin(2(k+1)\theta)}{2\sin\theta} \text{ as required.} \end{aligned}$$

$\therefore$ True for $n=k+1$ and for all n by PMI.

(ii). Using part (i) and the substitution $\theta = \frac{\pi}{2} - x$, or otherwise, show that

$$\sin x - \sin 3x + \sin 5x = \frac{\sin 6x}{2\cos x}$$

$$\sin x = \cos\left(\frac{\pi}{2} - x\right)$$

$$\cos(x \pm 2\pi) = \cos x \text{ and } \sin(x \pm 2\pi) = \sin x$$

$$\sin x - \sin 3x + \sin 5x = \sin\left(\frac{\pi}{2} - \theta\right) - \sin\left[3\left(\frac{\pi}{2} - \theta\right)\right] + \sin\left[5\left(\frac{\pi}{2} - \theta\right)\right]$$

$$= \cos\theta - \sin\left[\frac{\pi}{2} - (3\theta - \pi)\right] + \sin\left[\frac{\pi}{2} - (5\theta - 2\pi)\right]$$

$$= \cos\theta - \cos(3\theta - \pi) + \cos(5\theta - 2\pi) = \cos\theta - \cos(\pi - 3\theta) + \cos(5\theta) \quad \left[\begin{array}{l} \cos\theta \text{ is even;} \\ \text{period of } 2\pi \end{array}\right]$$

$$= \cos\theta + \cos(3\theta) + \cos(5\theta) = \frac{\sin 6\theta}{2\sin\theta} \quad [\text{From (i)}]$$

$$= \frac{\sin 6\left(\frac{\pi}{2} - x\right)}{2\sin\left(\frac{\pi}{2} - x\right)} = \frac{\sin(3\pi - 6x)}{2\cos x} = \frac{\sin(\pi + 2\pi - 6x)}{2\cos x}$$

$$= \frac{-\sin(2\pi - 6x)}{2\cos x} = \frac{-\sin(-6x)}{2\cos x} \quad [\sin\theta \text{ period } 2\pi]$$

$$= \frac{\sin(6x)}{2\cos x} \quad [\sin\theta \text{ is odd}]$$

8 Prove by mathematical induction that $2n(n-1)$ is divisible by 4 for all positive integers $n > 1$.

$$\text{For } n=2, 2n(n-1) = 4(1) = 4$$

which is divisible by 4

So true for $n=2$

Assume true for $n=k$

$$\text{That is, assume } 2k(k-1) = 4M, \quad \text{M an integer}$$

For $n=k+1$, it is required to show that

$$2(k+1)(k+1-1) = 4P, \quad P \text{ an integer}$$

$$\text{That is, } 2k(k+1) = 4P$$

$$LHS = 2k(k+1)$$

$$= 2k(k-1+2)$$

$$= 2k(k-1) + 4k$$

$$= 4M + 4k \quad (\text{by assumption})$$

$$= 4(M+k)$$

$$= 4P \text{ where } P = M+k \text{ is an integer.}$$

$$= RHS$$

So the result is true for $n=k+1$ if true for $n=k$. Hence the result is proven true by mathematical induction.

$$1^2 \times 2 + 2^2 \times 2^2 + 3^2 \times 2^3 + \dots + n^2 \times 2^n = (n^2 - 2n + 3)2^{n+1} - 6$$

$$n = 1$$

$$\text{LHS} = 1 \times 2 \quad \text{RHS} = (1 - 2 + 3)2 - 6$$

$$= 2 \quad = 2$$

$$\therefore \text{LHS} = \text{RHS}$$

so true for $n = 1$

Prove true for $n = k$

$$1^2 \times 2 + 2^2 \times 2^2 + \dots + k^2 \times 2^k = (k^2 - 2k + 3) \times 2^{k+1} - 6$$

Assume true for $n = k+1$

$$1^2 \times 2 + \dots + k^2 \times 2^k + (k+1)^2 \times 2^{k+1}$$

$$= ((k+1)^2 - 2(k+1) + 3) \times 2^{k+2} - 6$$

$$1^2 \times 2 + \dots + k^2 \times 2^k + (k+1)^2 \times 2^{k+1}$$

$$= (k^2 - 2k + 3)2^{k+1} - 6 + (k+1)^2 2^{k+1}$$

(using assumption)

$$= 2^{k+1}(k^2 - 2k + 3 + k^2 + 2k + 1) - 6$$

$$= 2^{k+1}(2k^2 + 4) - 6$$

$$= 2^{k+1}(k^2 + 2) - 6$$

$$\text{Now RHS} = (k^2 + 2k + 1 - 2k - 2 + 3)2^{k+2} - 6$$

$$= (k^2 + 2)2^{k+2} - 6$$

$$= \text{LHS}$$

$\therefore$ true by process of
mathematical induction
for $n \geq 1$.



Induction Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 Use mathematical induction to show that for all integers $n \geq 1$,

3

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Step 1: Let $n=1$

$$\text{LHS} = 1 \quad \text{RHS} = \frac{1(1+1)}{2}$$

$$= 1$$

$\therefore \text{LHS} = \text{RHS}$, true for $n=1$.

Step 2: Assume true for $n=k$, $k \geq 1$

$$1 + 2 + \dots + k = \frac{k(k+1)}{2}$$

✓ 1 mark
for step 1
and step 2

Step 3: Prove true for $n=k+1$

$$\begin{aligned} 1 + 2 + \dots + k + (k+1) &= \frac{(k+1)(k+1+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

$$\text{LHS} = 1 + 2 + \dots + k + (k+1)$$

$$= \frac{k(k+1)}{2} + (k+1) \quad (\text{from step 2 assumption})$$

✓ 1 mark for using
assumption

$$= \frac{k(k+1)}{2} + \frac{2(k+1)}{2}$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

✓ 1 mark
algebra

$$= \frac{(k+1)(k+2)}{2}$$

$$= \text{RHS}$$

Therefore, using the principle of mathematical induction,
the result is true for all $n \geq 1$.

- 2 Prove by mathematical induction that $2^{2n} + 6n - 1$ is divisible by 3 for all integers $n \geq 1$.

3

$$2^{2n} + 6n - 1 \text{ is divisible by } 3, n \geq 1$$

Step 1: Show true for $n=1$

$$2^{2(1)} + 6(1) - 1$$

$$= 9 \text{ which is divisible by } 3$$

Step 2: Assume true for $n=k$

$$\text{i.e. } 2^{2k} + 6k - 1 = 3P \text{ where } P \text{ is an integer}$$

$$2^{2k} = 3P - 6k + 1$$

Step 3: Prove true for $n=k+1$

$$\text{i.e. } 2^{2(k+1)} + 6(k+1) - 1 = 3Q \text{ where } Q \text{ is an integer.}$$

$$\text{LHS} = 2^{2(k+1)} + 6(k+1) - 1$$

$$= 2^{2k+2} + 6k + 6 - 1$$

$$= 2^2 \cdot 2^{2k} + 6k + 5$$

$$= 4(3P - 6k + 1) + 6k + 5 \quad (\text{from assumption})$$

$$= 12P - 24k + 4 + 6k + 5$$

$$= 12P - 18k + 9$$

$$= 3(4P - 6k + 3)$$

$$= 3Q$$

$$= \text{RHS}$$

$\therefore$ If the result is true for $n=k$, then
also true for $n=k+1$

Step 4: By the principle of mathematical
induction, the result is true for
all integers $n \geq 1$.

- 3 Use the mathematical induction to prove that $10^n + 3 \times 4^{n+2} + 5$ is divisible by 9, for all positive integers $n \geq 1$.

3

Step 1 - Base Case - prove true for $n=1$

$$\begin{aligned} \text{For } n=1 \quad & 10^n + 3 \times 4^{n+2} + 5 \\ &= 10^1 + 3 \times 4^{1+2} + 5 \\ &= 10 + 192 + 5 \\ &= 207 \\ &= 9 \times 23 \quad \text{Which is divisible by 9} \end{aligned}$$

$\therefore$ true for $n=1$.

Step 2 - Inductive hypothesis - assume true for $n=k$

$$\text{i.e. } 10^k + 3 \times 4^{k+2} + 5 = 9M \quad \text{where } M \in \mathbb{Z}$$

Step 3 - Inductive step - Prove true for $n=k+1$

Prove $10^{k+1} + 3 \times 4^{k+3} + 5$ is divisible by 9.

$$\begin{aligned} & 10^k \cdot 10 + 3 \cdot 4^{k+2} \cdot 4 + 5 \\ &= 10(9M - 3 \times 4^{k+2} - 5) + 3 \cdot 4^{k+2} \cdot 4 + 5 \\ &= 90M - 30 \times 4^{k+2} - 50 + 3 \cdot 4^{k+2} \cdot 4 + 5 \\ &= 90M - 30 \times 4^k \cdot 4^2 - 50 + 3 \cdot 4^k \cdot 4^3 + 5 \\ &= 90M - 480 \cdot 4^k - 45 + 192 \cdot 4^k \\ &= 90M - 288 \cdot 4^k - 45 \\ &= 9(10M - 32 \cdot 4^k - 5) \end{aligned}$$

Which is divisible by 9.

$\therefore$ true for $n=k+1$, since true for $n=k$

$\therefore$ By the principle of mathematical induction it is true for all integer $n \geq 1$.

- 4 Use Mathematical Induction to show that for all positive integers $n \geq 2$,

3

$$2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + n(n-1) = \frac{1}{3}n(n^2 - 1)$$

$$\text{RIP: } 2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + n(n-1) = \frac{1}{3}n(n^2 - 1), \quad n \geq 2$$

Step 1: For $n=2$

$$\text{LHS} = 2 \times 1 = 2$$

$$\text{RHS} = \frac{1}{3} \times 2 \times (2^2 - 1) = 2$$

$\therefore$ LHS = RHS, statement is true for $n=2$.

Step 2: Assume statement is true for $n=k$, $k \geq 2$ integer

$$\text{i.e. } 2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + k(k-1) = \frac{1}{3}k(k^2 - 1) \quad (*)$$

Step 3: Attempt to prove statement is true for $n=k+1$

$$\text{i.e. } 2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + (k+1)k = \frac{1}{3}(k+1)[(k+1)^2 - 1]$$

$$= \frac{1}{3}(k+1)(k^2 + 2k)$$

$$\text{LHS} = 2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + k(k-1) + (k+1)k$$

$$= \frac{1}{3}k(k^2 - 1) + (k+1)k \quad \text{from assumption } (*)$$

$$= \frac{1}{3}k(k+1)(k-1) + k(k+1)$$

$$= \frac{1}{3}(k+1)[k(k-1) + 3k]$$

$$= \frac{1}{3}(k+1)(k^2 - k + 3k)$$

$$= \frac{1}{3}(k+1)(k^2 + 2k)$$

$$= \text{RHS}$$

$\therefore$ Statement is true for $n=k+1$ if it is true for $n=k$.

Step 4 By the principle of mathematical induction, the statement is true for integers $n \geq 2$.

5 Prove by mathematical induction, that for all integers $n \geq 1$,

3

$$1 + 2 \times 2^{-1} + 3 \times 2^{-2} + 4 \times 2^{-3} + \dots + n \times 2^{-(n-1)} = \frac{2^{n+1} - n - 2}{2^{n-1}}$$

Step 1:

Proving the statement is true for $n = 1$ gives:

$$\begin{aligned} \text{LHS} &= 1 \times 2^{-(1-1)} \\ &= 1 \\ \text{RHS} &= \frac{2^{1+1} - 1 - 2}{2^{1-1}} \\ &= \frac{4 - 1 - 2}{1} \\ &= 1 \end{aligned}$$

LHS = RHS

Therefore, the statement is true for $n = 1$.

Step 2:

Assuming the statement is true for $n = k$ gives:

$$\begin{aligned} 1 + 2 \times 2^{-1} + 3 \times 2^{-2} + 4 \times 2^{-3} \\ + \dots + k \times 2^{-(k-1)} &= \frac{2^{k+1} - k - 2}{2^{k-1}} \end{aligned}$$

Step 3:

Proving that the statement is true for $n = k + 1$ gives:

$$\begin{aligned} 1 + 2 \times 2^{-1} + 3 \times 2^{-2} + 4 \times 2^{-3} + \dots + k \times 2^{-(k-1)} \\ + (k+1) \times 2^{-k} &= \frac{2^{k+2} - (k+1) - 2}{2^k} = \frac{2^{k+2} - k - 3}{2^k} \\ \text{LHS} &= 1 + 2 \times 2^{-1} + 3 \times 2^{-2} + 4 \times 2^{-3} + \dots \\ &\quad + k \times 2^{-(k-1)} + (k+1) \times 2^{-k} \\ &= \frac{2^{k+1} - k - 2}{2^{k-1}} + (k+1) \times 2^{-k} \quad (\text{by assumption}) \\ &= \frac{2^{k+1} - k - 2}{2^{k-1}} + \frac{k+1}{2^k} \\ &= \frac{2^{k+2} - 2k - 4}{2^k} + \frac{k+1}{2^k} \\ &= \frac{2^{k+2} - 2k - 4 + k + 1}{2^k} = \frac{2^{k+2} - k - 3}{2^k} = \text{RHS} \end{aligned}$$

If $n = k$ is true, then $n = k + 1$ is true. Therefore, by mathematical induction, the statement is true for $n \geq 1$.

6 Prove, by Mathematical Induction, that $5^{2n+1} + 2^{2n+1}$ is divisible by 7 for all integers $n \geq 0$.

3

$$\begin{aligned} \text{For } n = 0, \quad 5^{2n+1} + 2^{2n+1} &= 5 + 2 \\ &= 7 \quad \text{which is divisible by 7.} \end{aligned}$$

$\therefore$ Proven true for $n = 0$.

Assume true for $n = k$, i.e. $5^{2k+1} + 2^{2k+1}$ is divisible by 7.

i.e. $5^{2k+1} + 2^{2k+1} = 7P$ where $P \in \mathbb{R}$

Required to prove true for $n = k + 1$,

i.e. $5^{2(k+1)+1} + 2^{2(k+1)+1}$ is divisible by 7.

Proof:

$$\begin{aligned} 5^{2(k+1)+1} + 2^{2(k+1)+1} &= 5^{(2k+1)+2} + 2^{(2k+1)+2} \\ &= 5^2 \times 5^{2k+1} + 2^2 \times 2^{2k+1} \\ &= 25(7P - 2^{2k+1}) + 4 \times 2^{2k+1} \quad \text{by assumption} \\ &= 25 \times 7P - 25 \times 2^{2k+1} + 4 \times 2^{2k+1} \\ &= 25 \times 7P - 21 \times 2^{2k+1} \\ &= 7(25P - 3 \times 2^{2k+1}) \quad \text{which is divisible by 7.} \end{aligned}$$

$\therefore$ Proven true for $n = k + 1$.

If true for $n = k$, proven true for $n = k + 1$.

Since true for $n = 0$, true for $n = 0 + 1$, $n = 1 + 1$, ...,

therefore, true for all integer $n (n \geq 0)$.

7 (i). Show that $\frac{1}{(n+1)!} - \frac{n+1}{(n+2)!} = \frac{1}{(n+2)!}$ 1

$$\text{LHS} = \frac{1}{(n+1)!} - \frac{n+1}{(n+2)!} = \frac{n+2}{(n+2)(n+1)!} - \frac{n+1}{(n+2)!} = \frac{n+2-(n+1)}{(n+2)!} = \frac{1}{(n+2)!} = \text{RHS}$$

(ii). Use Mathematical Induction to prove that, for all integers $n \geq 1$, 3

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$$

ii) Let $P(n)$ be given proposition.

Base case : Prove true for $n=1$

$$\text{LHS} = \frac{1}{(1+1)!} = \frac{1}{2} \quad \text{RHS} = 1 - \frac{1}{(1+1)!} = \frac{1}{2}$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore P(1)$ is true.

Assume true for $n=k$;

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$$

Prove true for $n=k+1$;

$$\text{RTP: } \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} = 1 - \frac{1}{(k+2)!}$$

$$\text{LHS} = 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \quad (\text{By assumption})$$

$$= 1 - \left(\frac{1}{(k+1)!} - \frac{k+1}{(k+2)!} \right)$$

$$= 1 - \frac{1}{(k+2)!} \quad [\text{from i)]}$$

$$= \text{RHS}$$

$\therefore P(k+1)$ is true for all $n \in \mathbb{Z}^+$

$\therefore$ True by Mathematical Induction.

8 Use the principle of mathematical induction to show that for all integers $n \geq 1$ 3

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$$

When $n = 1$;

$$\text{LHS} = 1^3 = 1$$

$$\begin{aligned} \text{RHS} &= \frac{1}{4}(1)^2(1+1)^2 \\ &= \frac{1}{4} \times 1 \times 4 \\ &= 1 \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

Hence the result is true for $n=1$

Assume the result is true for $n = k$, where $k \in \mathbb{Z}^+$

$$\text{i.e. } 1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{1}{4}k^2(k+1)^2$$

Prove the result is true for $n = k+1$

$$\text{i.e. } 1^3 + 2^3 + 3^3 + \dots + (k+1)^3 = \frac{1}{4}(k+1)^2(k+2)^2$$

PROOF:

$$\begin{aligned} &1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 \\ &= \frac{1}{4}k^2(k+1)^2 + (k+1)^3 \quad (\text{by assumption}) \\ &= \frac{1}{4}(k+1)^2[k^2 + 4(k+1)] \\ &= \frac{1}{4}(k+1)^2(k^2 + 4k + 4) \\ &= \frac{1}{4}(k+1)^2(k+2)^2 \end{aligned}$$

Hence the result is true for $n = k+1$, if it is true for $n = k$

Since the result is true for $n = 1$, then it is true $\forall n \in \mathbb{Z}^+$ by induction.

9 Prove by mathematical induction that $n^3 + 2n$ is divisible by 3 for all positive integers n .

2

- Test for $n=1$
 $1^3 + 2(1) = 3$ which is divisible by 3
 $\therefore$ The statement is true for $n=1$
- Assume the statement is true for $n=k$
i.e. $k^3 + 2k = 3P$ for integer P
- Prove for $n=k+1$
i.e. $(k+1)^3 + 2(k+1) = 3Q \quad Q \in \mathbb{Z}$

$$\begin{aligned} \text{LHS} &= (k+1)^3 + 2(k+1) \\ &= (k^3 + 3k^2 + 3k + 1) + (2k + 2) \\ &= (k^3 + 2k) + 3(k^2 + k + 1) \\ &= 3P + 3(k^2 + k + 1) \quad \text{from assumption} \\ &= 3(P + k^2 + k + 1) \\ &= 3Q, \quad Q = P + k^2 + k + 1 \in \mathbb{Z} \end{aligned}$$
 $\therefore$ The statement is true for $n=k+1$, if it is true for $n=k$.
- By mathematical induction, it is true for all positive integers.

10 (i). Prove that $n + 1$ is a factor of $P(n) = 4n^3 + 18n^2 + 23n + 9$.

1

$$P(n) = 4n^3 + 18n^2 + 23n + 9$$
$$P(-1) = -4 + 18 - 23 + 9 = 0$$

$n + 1$ is a factor of $P(n)$

(ii). Hence, use mathematical induction to prove that for all integers $n \geq 1$,

$$1 \times 3 + 3 \times 5 + 5 \times 7 + \cdots + (2n+1)(2n-1) = \frac{n(4n^2 + 6n - 1)}{3}$$

When $n = 1$,

$$LHS = 1 \times 3 = 3 \quad RHS = \frac{1(4 \times 1^2 + 6 \times 1 - 1)}{3} = 3 = LHS$$

The statement is true for $n = 1$.

Assume the statement is true for $n = k$, where k is an integer $k \geq 1$

$$1 \times 3 + 3 \times 5 + 5 \times 7 + \cdots + (2k+1)(2k-1) = \frac{k(4k^2 + 6k - 1)}{3}$$

Prove the statement is true for $n = k + 1$

$$= \frac{(k+1)(4k^2+14k+9)}{3}$$

Proof:

$$LHS = 1 \times 3 + 3 \times 5 + 5 \times 7 + \cdots + (2k+1)(2k-1) + (2k+3)(2k+1)$$

$$LHS = \frac{k(4k^2 + 6k - 1)}{3} + (2k + 3)(2k + 1)$$

$$LHS = \frac{k(4k^2 + 6k - 1) + 3(2k + 3)(2k + 1)}{3}$$

$$LHS = \frac{4k^3 + 6k^2 - k + 3(4k^2 + 8k + 3)}{3}$$

$$LHS = \frac{4k^3 + 6k^2 - k + 12k^2 + 24k + 9}{3}$$

$$LHS = \frac{4k^3 + 18k^2 + 23k + 9}{3}$$

Let $P(k) = 4k^3 + 18k^2 + 23k + 9$

From part (i), $k + 1$ is a factor of $P(k)$

$$\text{Now, } LHS = \frac{4k^3 + 18k^2 + 23k + 9}{3}$$

$$\begin{array}{r}
 \overline{4k^2 + 14k + 9} \\
 k+1 \overline{) 4k^3 + 18k^2 + 23k + 9} \\
 \underline{4k^3 + 4k^2} \\
 14k^2 + 23k + 9 \\
 \underline{ 14k^2 + 14k} \\
 9k + 9 \\
 \underline{ 9k + 9} \\
 0
 \end{array}$$